$$r_{\mathrm{m}} = \frac{1+\sqrt{5}}{4} d \ . \tag{8}$$

Q_0 Q_{10}

$$\frac{Q_0}{T_0} - \frac{Q_{10}}{T_1} \le 0 \ . \tag{9}$$

Q_2

Q_{12}

$$-\frac{Q_{12}}{T_1} + \frac{Q_2}{T_2} \le 0 \ . \tag{10}$$

$$Q_1 = Q_{10} + Q_{12} \ . \tag{11}$$

(9)、(10)

$$\frac{Q_0}{T_0} + \frac{Q_2}{T_2} - \frac{Q_1}{T_1} \le 0 \ . \tag{12}$$

$$Q_1 = Q_0 + Q_2 \ . \tag{13}$$

Q_1

$$Q_0' = Q_1 \ . \tag{14}$$

(12)、(13)

$$Q_0 \ge \frac{T_0 \left(T_1 - T_2 \right)}{T_1 \left(T_0 - T_2 \right)} Q_1 \ . \tag{15}$$

$$\frac{Q_0' - Q_0}{Q_0'} \le \frac{T_2 \left(T_0 - T_1 \right)}{T_1 \left(T_0 - T_2 \right)} \ . \tag{16}$$